

Umeshkumar Ghaskata

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University at Buffalo, SUNY (MS in Computer Science)

May 2026

Indian Institute of Technology, Bombay (B.Tech in Aerospace Engineering)

May 2024

PROFESSIONAL EXPERIENCE

DRONES Lab (NASA AIST-21 Project) — ML Engineer *Carr Astronautics, UB, USA* [Aug'2025-Jan'2026]

- Designed, trained, evaluated, and deployed **DL models** (PCA+MLP, FPGA-based architectures) for hyperspectral image classification improving land-cover recognition accuracy from 82.1% to 92.2%.
- Reduced **edge inference latency** from **2.7s to 32ms** using optimized FPGA Model for real-time output.
- Built cross-platform benchmarking framework across **RTX 4060, Jetson Orin, and Xavier** to quantify deployment trade-offs and guide edge hardware selection.

Roswell Park Research Corporation — Full Stack Developer *Buffalo, USA* [Jan'2026-May'2026]

- Fine-tuned the Gemma 3-1B model on **AWS EC2 Tesla 4 GPU** using QLoRA adapter and **Hugging Face PEFT** and deployed to **UB server** to build a personalized conversational AI.
- Developed cross-platform mobile applications using **React Native 0.78**, implementing multi-screen navigation, daily check-ins, push notifications, and local storage support for both **iOS and Android**.
- Engineered **CI/CD pipelines** reducing deployment time by **40%**, and optimized **Redux state management** to reduce redundant API calls by **25%**.

Scitara Corporation — Software Developer Intern *Mumbai, India* [Dec'2022-Feb'2023]

- Developed and tested mock **RESTful API integrations**, creating virtual server responses that reduced manual testing effort by **30%** and accelerated software validation by waiving credentials requirement.

PROJECTS

Multimodal AI Platform with RAG, Streaming [GitHub](#)  [Jan'2026–Mar'2026]

Transformers, LangChain, FAISS, CLIP, Kafka, PySpark, MLflow, FastAPI, Docker, AWS/GCP

- Built a **multimodal AI system** integrating **RAG pipelines**, **Agentic Workflow**, real-time streaming (Kafka), and big data processing (PySpark) for context-aware reasoning across text, documents, and images
- Designed an **end-to-end retrieval and inference pipeline** using CLIP embeddings, FAISS vector search, and LLMs with re-ranking, implementing **MLflow-based** model lifecycle management with automated retraining triggered by data and model drift

Deepface-Similarity-Engine | Image Processing [GitHub](#)  [Jul'2025–Sep'2025]

TensorFlow, Keras, OpenCV, MTCNN, VGGFace (ResNet50), NumPy, Streamlit

- Built a facial recognition system using **MTCNN** for face detection and **VGGFace (ResNet50)** for feature extraction, generating 2048-d embeddings for identity classification
- Developed an end-to-end pipeline with **TensorFlow** and **NumPy**, and deployed an interactive interface using **Streamlit** for real-time image upload and inference

Google-Drive-Clone | Full-Stack Web Application [GitHub](#)  [May'2024–Aug'2024]

React.js, Firebase Auth, Cloud Firestore, Firebase Storage, Material UI, JavaScript, CSS

- Developed a fully functional **Google Drive clone** using **React.js** and **Firebase**, implementing Google Sign-In authentication, real-time file management with **Cloud Firestore**.
- Built a responsive file management dashboard with reusable React components, dynamic file rendering, upload modals, and formatted file metadata display using **Material UI** and Firebase

TECHNICAL SKILLS SUMMARY

Languages	Python, C++, JavaScript, SQL, React-Native
ML/AI Framework	PyTorch, NLTK, Scipy, TensorFlow, Scikit-learn, Transformers, LangChain, LlamaIndex, RAG, LLM Fine-Tuning, PySpark, SFT, XGBoost
Software Engineering	Object-Oriented Programming (OOP), Data Structures & Algorithms, Design Patterns, REST APIs, Microservices, Distributed Systems, SDLC, Agile Development, Debugging, Unit Testing, Code Reviews, Git, GitHub
Cloud & DB	AWS, Docker, Redis, FAISS, PostgreSQL, MySQL, Firebase, Distributed Systems, CI/CD
Certifications	Generative AI Leader(Google)

SCHOLASTIC ACHIVEMENT

- Ranked in the **top 0.7% of 1.3+ million candidates** in national Joint Entrance Examination [2020]